

Bangladesh University of Engineering and Technology

Course no: EEE 208

Course title: Electronic Circuit-II Laboratory(Software)

Project Title: **Logarithmic Amplifier Using matched transistor**

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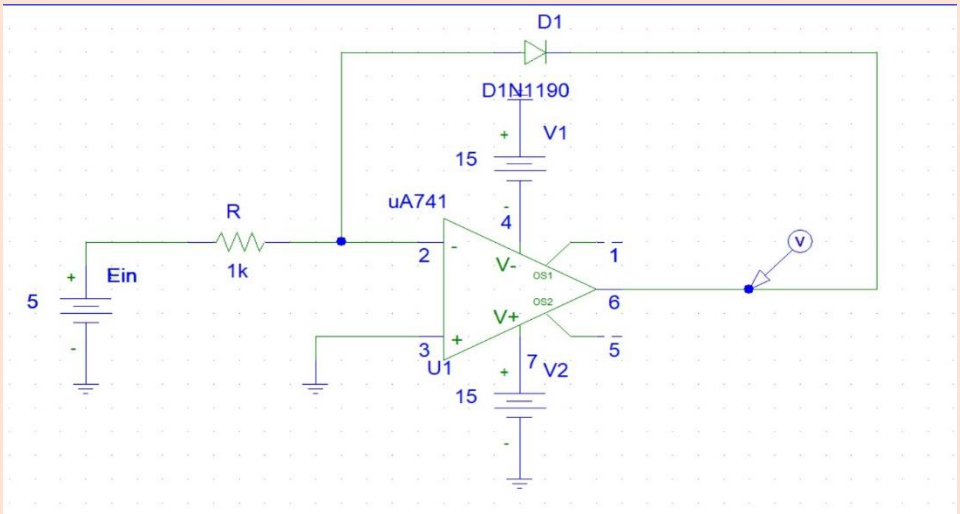
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Documentation

Logarithmic Amplifier Using matched transistor

Objective: We may need to use amplifier circuits for various reason in electrical and electronics industry. Logarithmic amplifier is one of them. It's an essential part of analog computer. Apart from performing mathematical expressions, logarithmic amplifier is useful for signal compression,reduce or enhance dynamic range of the signal, linearize the signal and so on. If we want to construct a general logarithmic amplifier circuit, we may design like this



Current through diode,

$$I_D = I_S e^{-\frac{V_o}{nV_T}}$$

Pin 3 is grounded.

Considering virtual ground,

current through resistance,

$$I_R = \frac{E_i}{R}$$

So, $I_D = I_R$

$$\Rightarrow I_S e^{-\frac{V_o}{nV_T}} = \frac{E_i}{R}$$

$$\Rightarrow -\frac{V_o}{nV_T} = \log\left(\frac{E_i}{RI_S}\right)$$

$$\Rightarrow V_o = -nV_T \log\left(\frac{E_i}{RI_S}\right)$$

So, the expression for the output voltage of a conventional logarithmic amplifier using diode is,

$V_o = -nV_T \ln\left(\frac{Vin}{R\,Is}\right)$

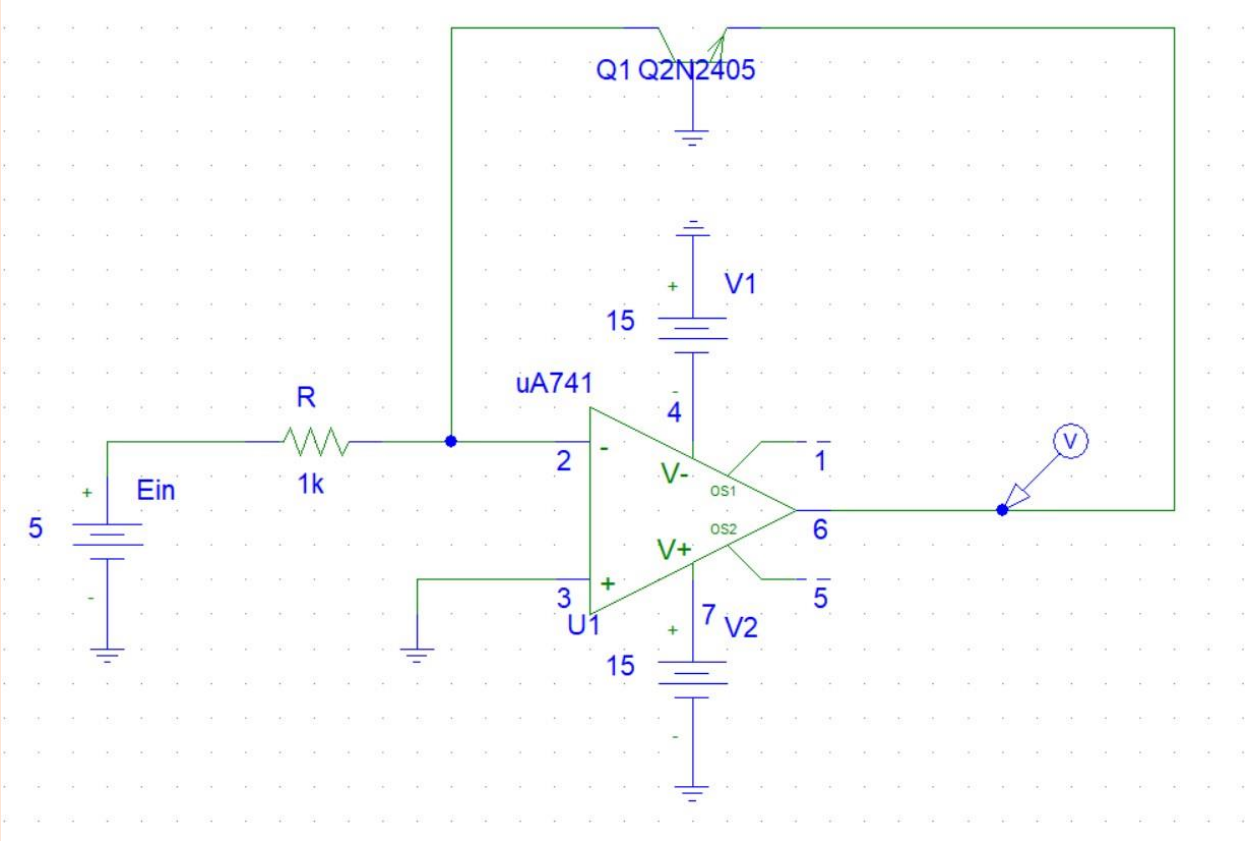
This amplifier is used when it is desired to have the output voltage

proportional to the logarithm of the input voltage.

Thus we can see, Vo depends on the

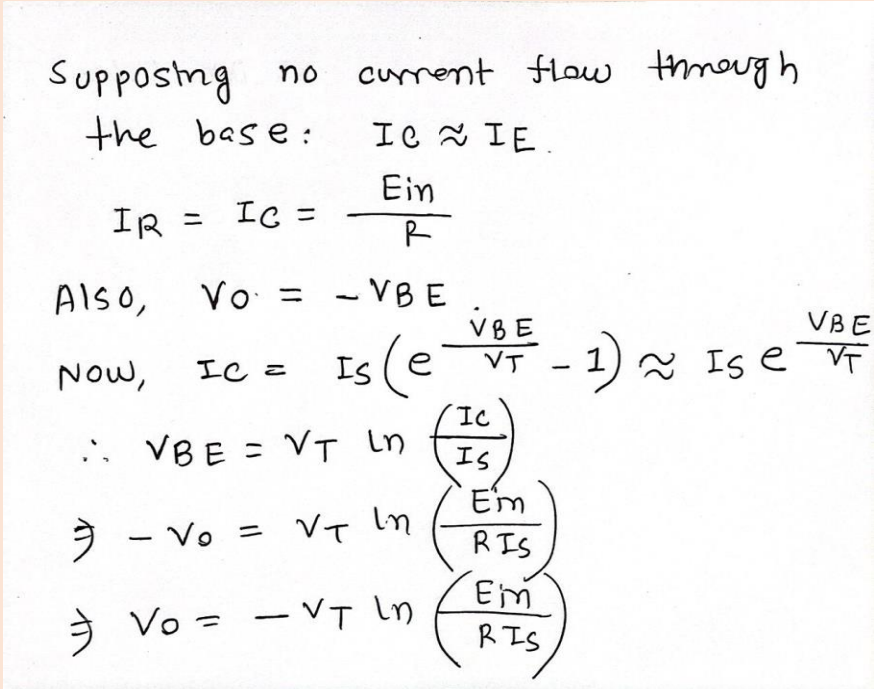
i) Ideality factor,n
ii) Thermal voltage V_T
iii) Reverse saturation current, I_S

Here, V_T and I_s are temperature dependent parameters. With the change of temperature they will change gradually. Ideality factor n is used to be in the range of 1 and 2. Because of the ideality factor, the dynamic range of circuit used to get reduced,which is one of the disadvantages of using diode based logarithmic amplifier. Here if we can use a transistor , the problem is expected to be solved.



In this circuit, V_B = 0V, V_C= 0V considering virtual ground.

So, only base emitter junction is in forward bias and so effectively this transistor will act as a diode.

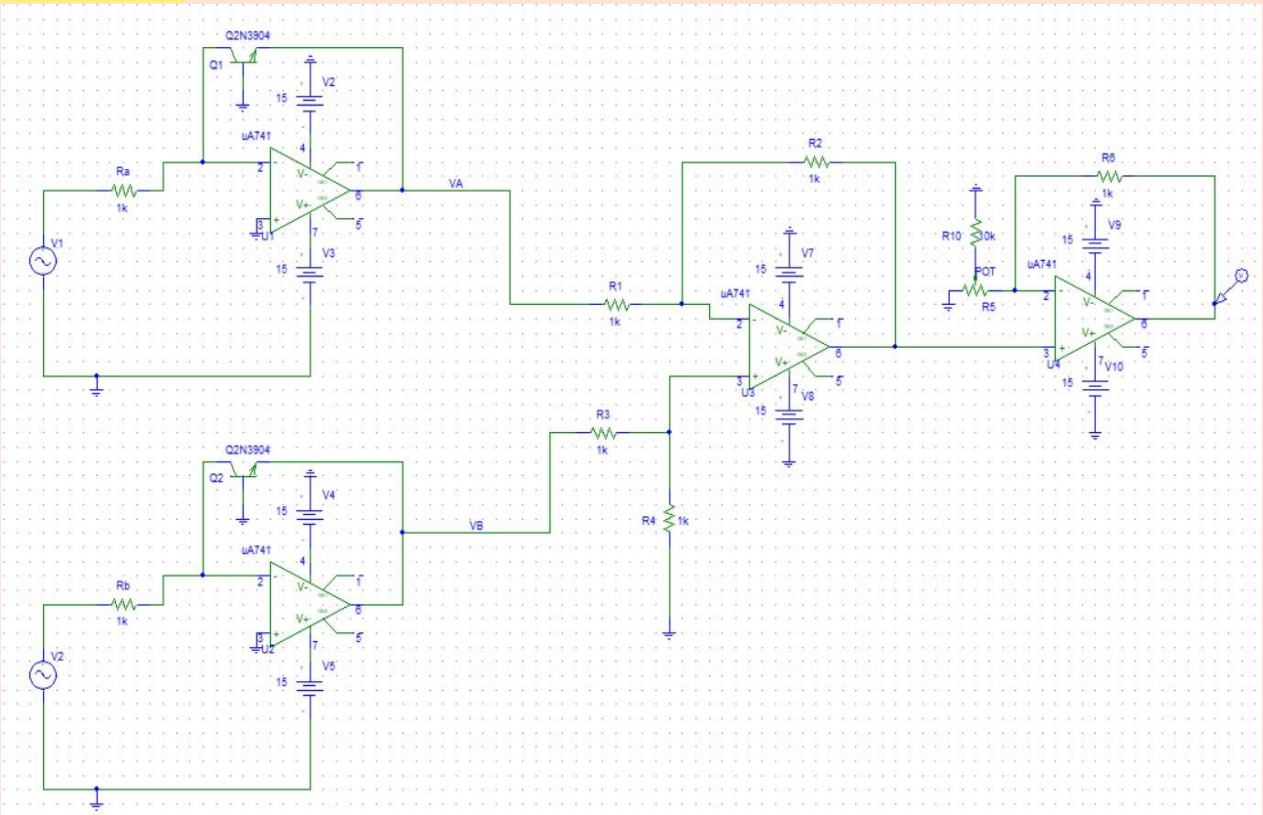


So, the expression for the output voltage of a transistor-based logarithmic amplifier using diode is,

$V_o = -V_T \ln\left(\frac{Vin}{R\,Is}\right)$

Where the ideality factor ,n =1, thus the dynamic range of the circuit is increased. But a problem still arises that, with temperature, Vo will change gradually. So, Instead of diodes, it is possible to use a matched pair of transistors connected to remove from the expression for n, the factor whose value normally depends on the current flowing through the diode.So we need to find a temperature compensated logarithmic amplifier with matched transistors so that we can nullify the effect of V_T with a component. We can use a temperature dependent resistor in that case.

Temperature compensated logarithmic amplifier with matched transistors:



Here,

$$V_A = -V_T \ln\left(\frac{V_1}{R_a I_S}\right)$$

$$V_B = -V_T \ln\left(\frac{V_2}{R_b I_S}\right)$$

Suppose, $R_a = R_b = R$

$$V_B - V_A = V_T \ln\left(\frac{V_1}{RI_S}\right) - V_T \ln\left(\frac{V_2}{RI_S}\right)$$

$$= V_T \ln\left(\frac{V_1}{RI_S} \times \frac{RI_S}{V_2}\right)$$

$$= V_T \ln\left(\frac{V_1}{V_2}\right)$$

Here the output voltage expression is independent of Is.

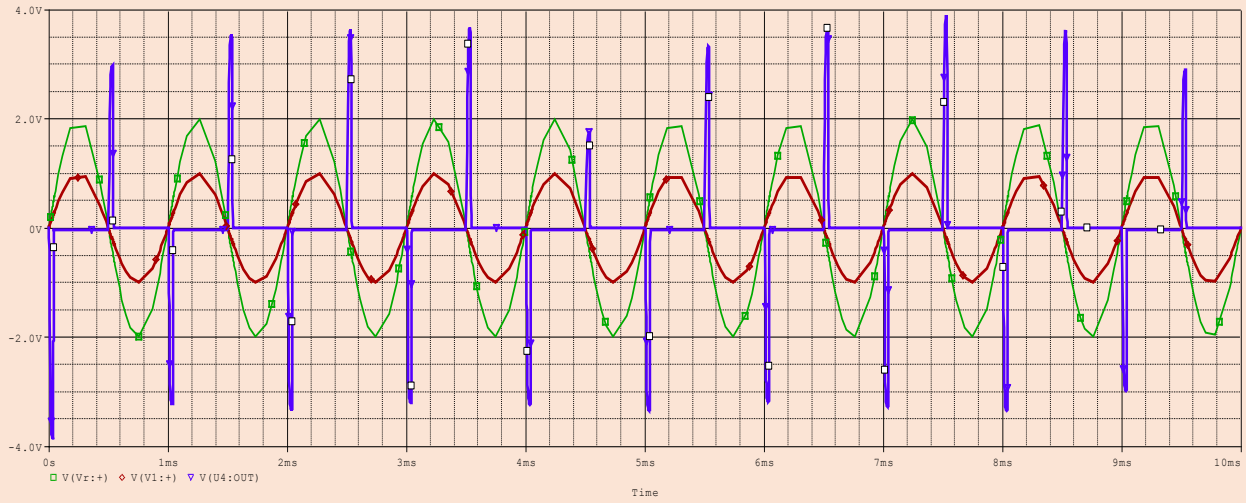
If we set the R_1, R_2, R_3 and R_4 value such that $\frac{R_1}{R_2} = \frac{R_3}{R_4}$, we can write, the output of the summing amplifier $= V_B - V_A$.

$$\therefore V_o = \left(1 + \frac{R_6}{R_5}\right)(V_B - V_A)$$

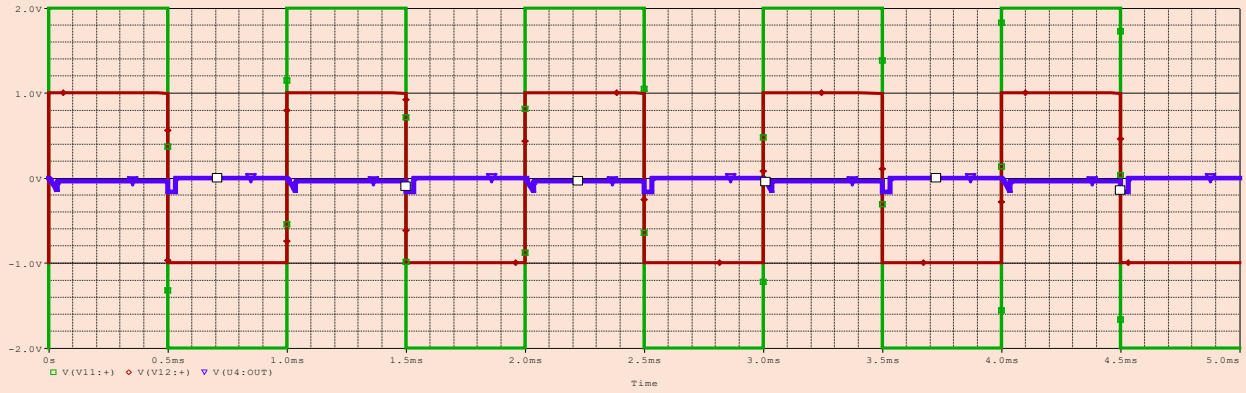
$$= \left(1 + \frac{R_6}{R_5}\right) V_T \ln\left(\frac{V_1}{V_2}\right).$$

Here, R_5 is a temperature dependent resistance. If we select R_5 such a way that it can nullify the effect of V_T and thus the whole circuit becomes completely temperature independent.

Output of the circuit for sine wave input:



Output of the circuit for sine wave input:

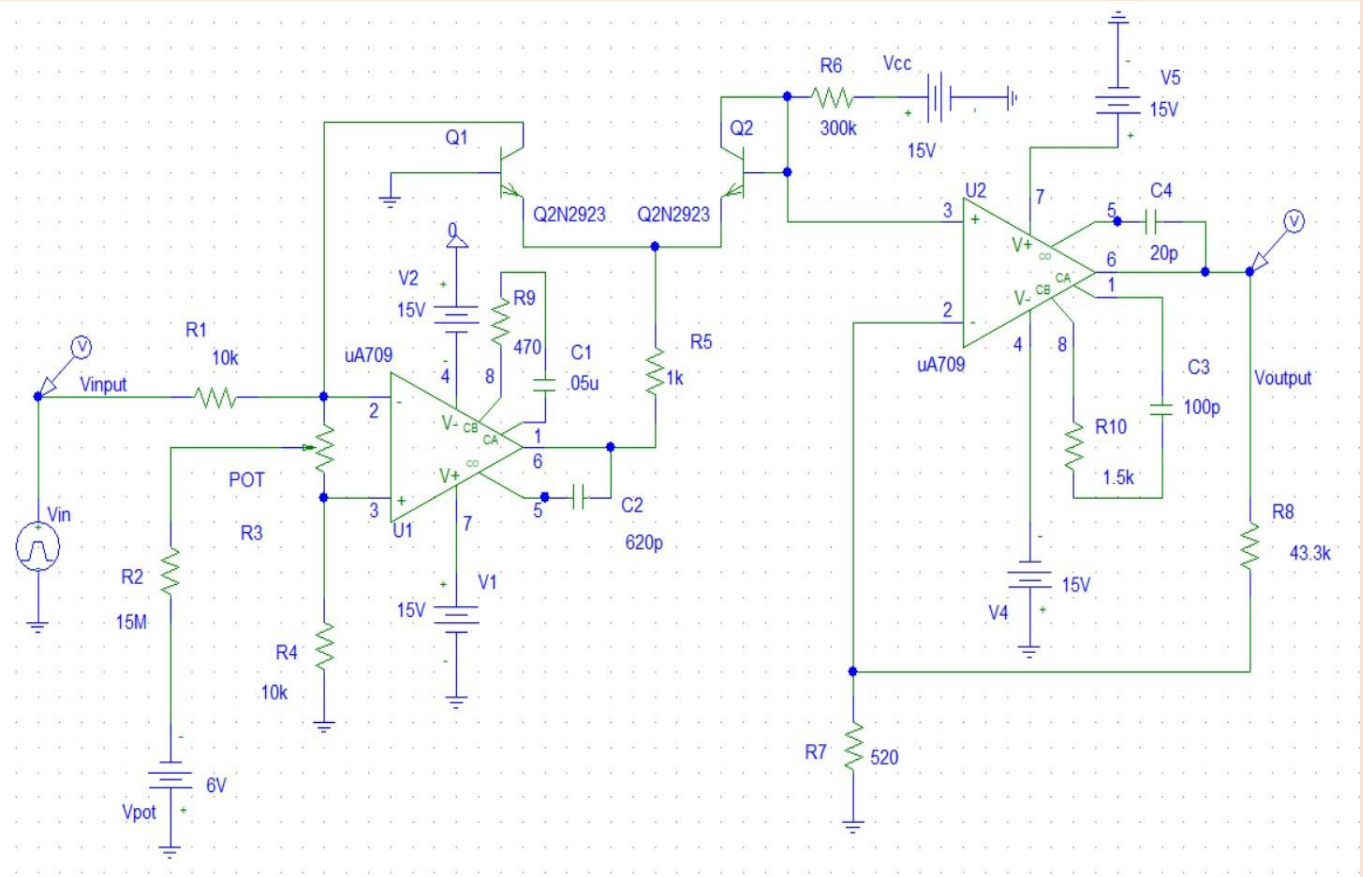


So our concern is to construct a circuit such a way that it becomes temperature independent, transistor based and stable enough to accomplish the amplification.

Name of the software used: PSpice version 9.2

Specification of the parts/models : i) Op-Amp model uA709
ii) Transistor model Q2N2923
iii)DC voltage source for biasing
iv)Potentiometre
v)Input voltage for Vsin, Vpulse
vi) Required resistor & capacitors, R & C

Circuit diagram:



Operation of the circuit:

Q1 is used as the feedback element around the first OP AMP. If we neglect V_{BE1}- V_{BE2} with respect to Vcc, and since I_{B2} « I_{c2}, then

$I_{C2} \approx \frac{Vcc}{R6}$ and $I_{C1} = \frac{Vs}{R1+R4} = \frac{Vs}{2R1}$

Now we know, $V_{BE1} - V_{BE2} = V_T \ln I_{C1}\ V_T \ln I_{C2} = V_T \ln (\frac{Vs}{2R1}.\frac{R6}{Vcc})$

Since the base of Q1 is grounded, the negative of the above voltage appears at the non-inverting terminal of the second operational amplifier, whose gain is determined by resistors R7 and Rs. Hence

$$\mathbf{V_o = -VT\ \frac{R7+Rs}{R7}\ ln\ (\frac{Vs}{2R1}.\frac{R6}{Vcc})}$$

The transfer function of the amplifier is plotted below for various operating temperatures. It is seen that the dynamic range extends over 5 mV to 50 V of input voltage, or 80 dB.

$\frac{d\,Vo}{d(\ln Vs)}$ $-V_T\frac{R7+Rs}{R7} = -0.026\times\frac{43.8}{0.52} = -2.20$

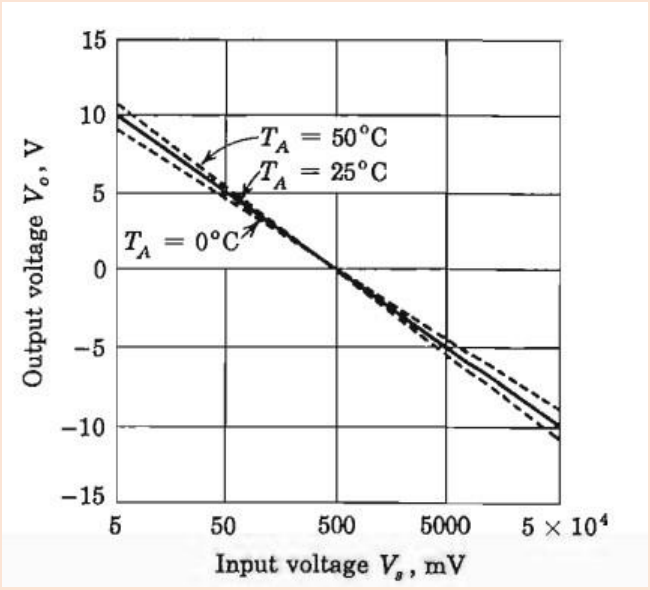


Fig : Transfer characteristics of logarithmic amplifier with matched transistors

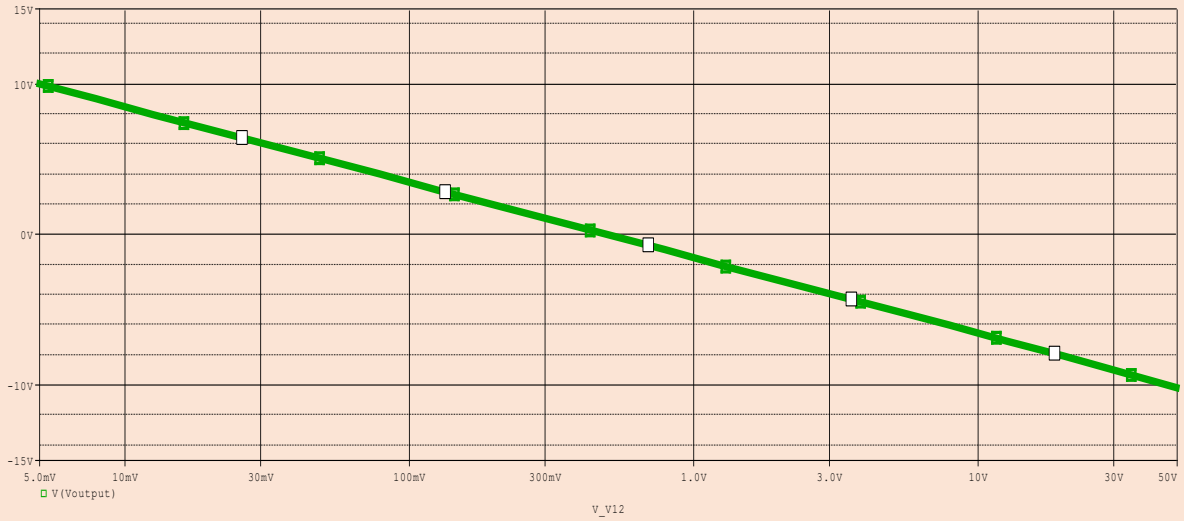
Ref:Integrated electronics- Millman and Halkias

(Courtesy of Fairchild Semiconductor, Inc.)

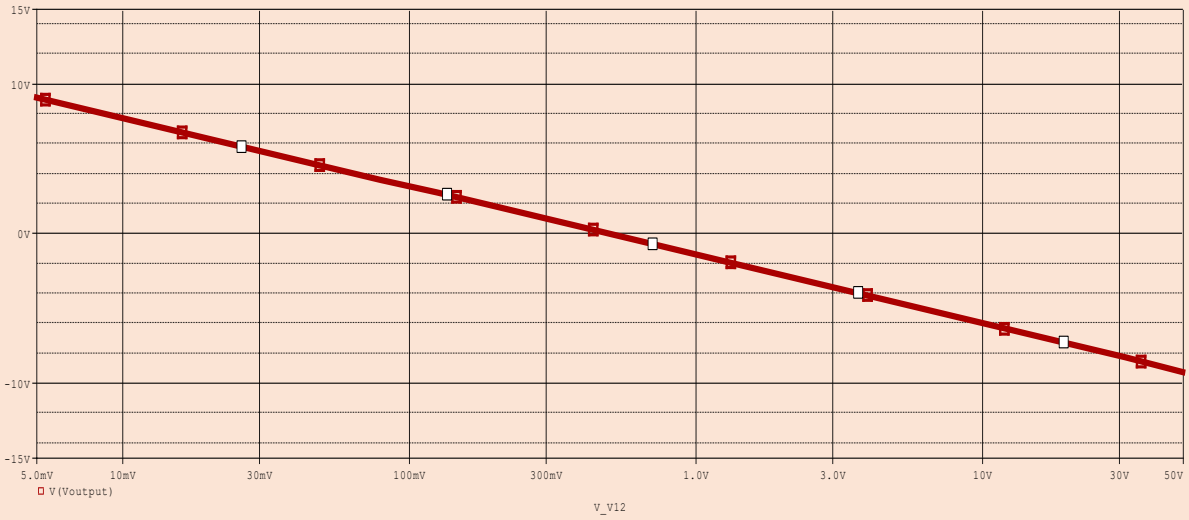
Some output vs input waveforms:

Transfer characteristics by varying temperature**:**

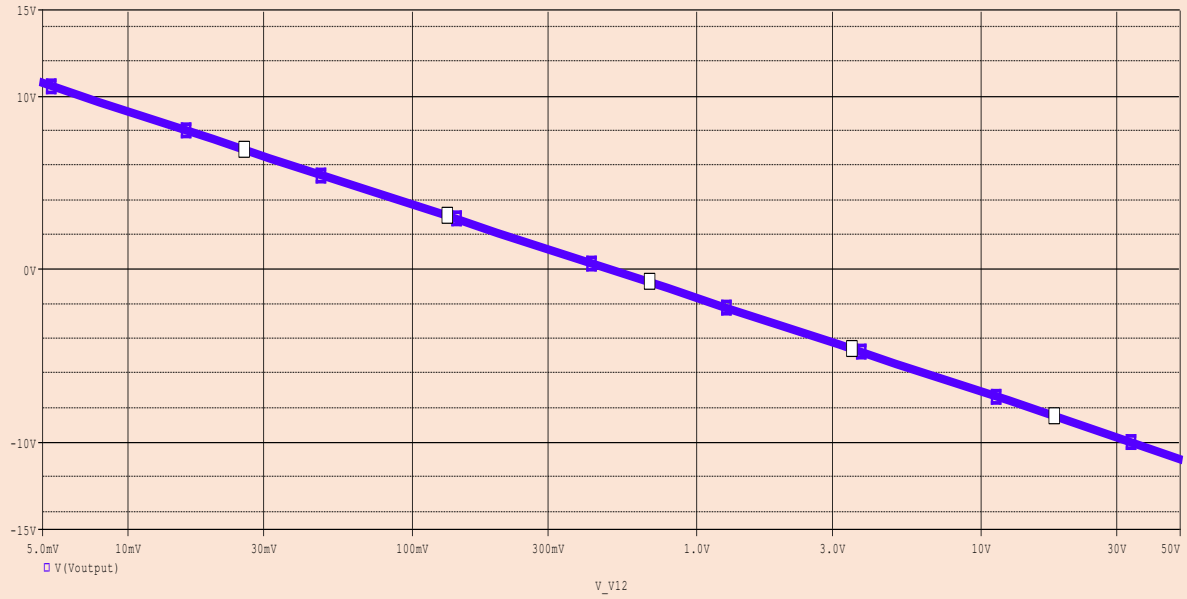
For T_A = 25º C:



For T_A =0ºC:

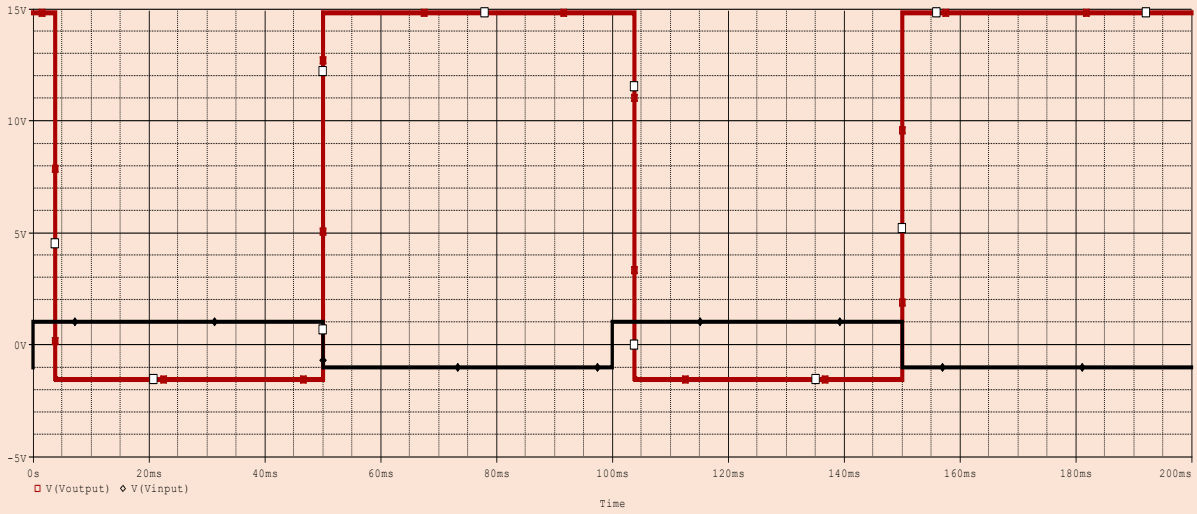


For T_A = 50°C:

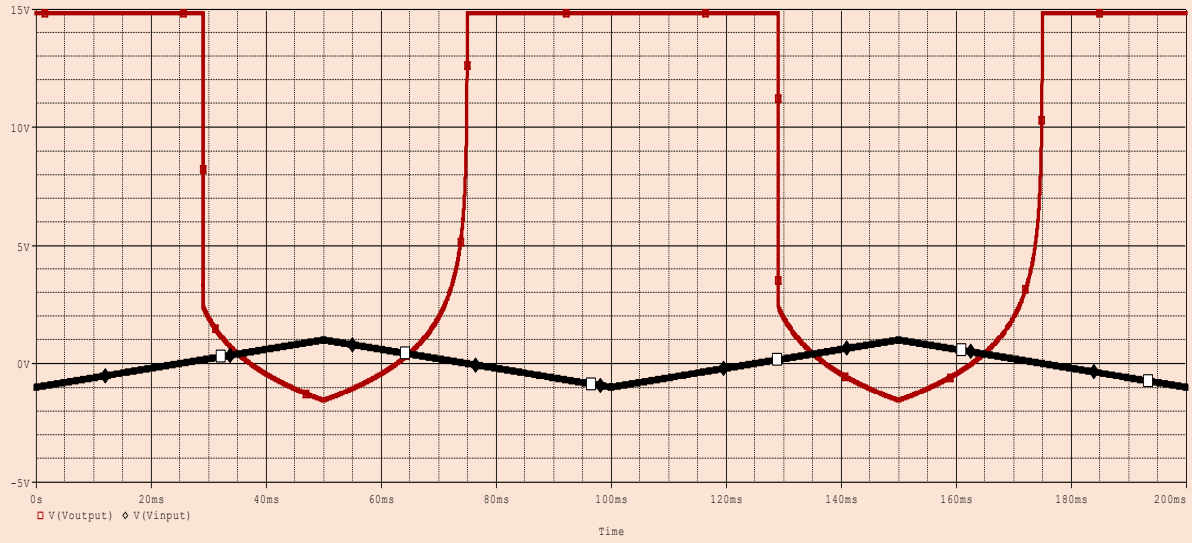


Output by changing time dependent input:

For Vin = 2V p-p square wave:



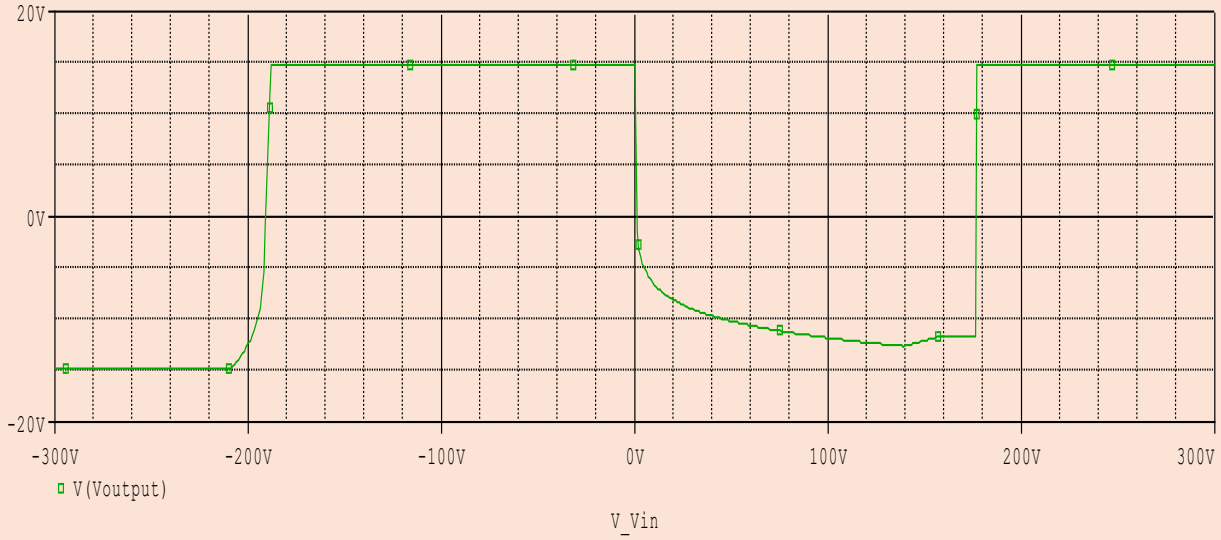
For Vin = 2V p-p trangular wave:



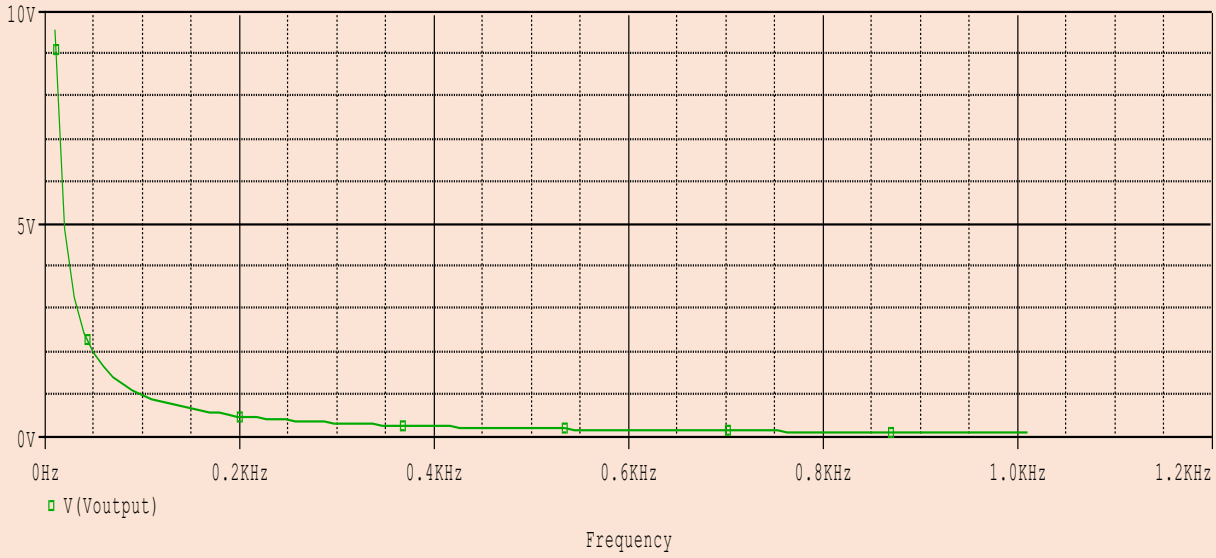
For Vin = 2V p-p sine wave:



Output in DC sweeping:



Output in AC sweeping:



This circuit also restricts high frequency input wave.

Application of the project:

Logarithmic amplifiers can be used in the following fields:

1.Audio Compression: Logarithmic amplifiers are commonly used in audio processing to implement compression and automatic gain control

(AGC) circuits. In audio, our perception of loudness is logarithmic, so logarithmic amplifiers help to process audio signals in a way that matches our hearing sensitivity.

2.Signal Strength Measurement: Logarithmic amplifiers are used in RF (radio frequency) and microwave systems to measure and display signal strength on devices like signal strength meters or on-screen indicators. They convert the wide range of RF signal strengths into a more manageable and linear representation.

3.Photodiode Amplification: In photodetection applications, where the input signal is generated by light hitting a photodiode, logarithmic amplifiers can be used to linearize the response of the photodiode to light intensity. This is useful in applications such as light detection and ranging (LiDAR) systems.

4.Decibel (dB) Measurement: Logarithmic amplifiers are integral to dB measurement systems. The decibel scale, used to express signal power ratios, is logarithmic in nature. Logarithmic amplifiers are employed to convert linear signal levels into dB scales.

5.Chemical Analysis: In certain chemical analysis methods, such as pH measurement using ion-selective electrodes, logarithmic amplifiers are used to convert the output of these electrodes into a pH value due to the logarithmic nature of pH scales.

6.Geiger-Muller Counters: Logarithmic amplifiers are used in radiation detection devices like Geiger-Muller counters to convert the detected radiation pulses into a logarithmic scale, making it easier to interpret and analyze radiation levels.

7.Environmental Monitoring: Logarithmic amplifiers can be used in environmental monitoring systems to process sensor data related to parameters like air quality, pollution levels, and radiation exposure.

8.Video and Imaging: Logarithmic amplifiers find use in certain video and imaging applications where capturing a wide range of light intensities.

9.Satellite Communication: Logarithmic amplifiers are employed in satellite communication systems to process weak signals from distant satellites and amplify them in a way that matches the logarithmic scale of signal strength.

These examples further illustrate how logarithmic amplifiers are crucial in various scientific, industrial, and technological applications .

In our project, we used matched transistor for logarithmic amplifier design.

Advantages of using matched transistors:

Using matched transistors in a logarithmic amplifier offers several advantages due to the precision and consistency that comes from the matching process:

- Improved Accuracy: Matched transistors have very similar electrical characteristics, such as threshold voltage, transconductance, and saturation current. This results in a more accurate and consistent logarithmic response across different circuits and devices.
- Reduced Error and Drift: The closely matched transistors help minimize errors and drift caused by variations in transistor characteristics. This is crucial in applications where accurate logarithmic conversion is required over time and temperature changes.
- Linearity: Matched transistors can provide better linearity in the logarithmic response, ensuring that the output accurately represents the input signal's logarithmic relationship.
- Reduced Nonlinearity Distortion: Using matched transistors can mitigate nonlinearity distortion caused by mismatched components. Nonlinearity distortion can lead to signal compression or expansion, affecting the accuracy of the logarithmic conversion.
- Higher Dynamic Range: Matched transistors can extend the dynamic range of the logarithmic amplifier. This is particularly useful when dealing with input signals that span a wide range of amplitudes, as matched transistors help maintain accuracy across this range.
- Consistency Across Multiple Devices: When manufacturing logarithmic amplifiers in large quantities, using matched transistors ensures that the performance characteristics are consistent across different devices.

- Higher Signal-to-Noise Ratio (SNR): The accuracy and linearity provided by matched transistors can contribute to a higher SNR in the output signal, improving the quality of the amplified logarithmic signal.

It's important to note that while using matched transistors offers these advantages, it also requires careful manufacturing processes and testing to ensure the desired matching is achieved. Additionally, matched transistors might come at a higher cost compared to regular transistors due to the stringent matching requirements during production. Here we just used software simulation for this circuit. But in practical case, we must be aware of cost and other things.

Discussion on Limitation of design:

Designing a logarithmic amplifier with matched transistors can be a complex task, and there are several limitations and challenges that designers need to be aware of. Here are some common limitations of such designs:

- ❖ Transistor Matching: While using matched transistors is essential for accuracy, finding or manufacturing perfectly matched transistors can be difficult and expensive. There may still be some degree of mismatch, which can lead to errors in the logarithmic response.

- ❖ Temperature Sensitivity: Transistor characteristics, including threshold voltage and transconductance, can vary with temperature. This sensitivity to temperature can introduce errors

in the logarithmic response, especially in applications with wide temperature ranges.

- Power Supply Sensitivity: The performance of logarithmic amplifiers can be sensitive to variations in the power supply voltage. Any fluctuations in the supply voltage can affect the linearity and accuracy of the circuit.

- Noise: Transistors are inherently noisy devices. Noise can be amplified along with the signal, leading to a degraded signal-to-noise ratio (SNR). Careful circuit design and filtering techniques are necessary to minimize noise contributions.

- Limited Dynamic Range: Logarithmic amplifiers have a limited dynamic range. They work well for signals within a certain input voltage range, but they may not accurately process signals that fall outside this range. This limitation can be a significant drawback in some applications.

- Frequency Response: The frequency response of logarithmic amplifiers may not be uniform across a wide range of frequencies. Depending on the design, there may be variations in the gain and phase response, which can impact the fidelity of the output signal.

- Nonlinearity at Extremes: Logarithmic amplifiers can exhibit nonlinearity, especially at the extreme ends of the input voltage

range. This nonlinearity can introduce errors in applications where high accuracy is required.

- Calibration Challenges: Calibrating a logarithmic amplifier can be a complex and time-consuming process. Achieving precise calibration over all operating conditions can be challenging, and it may require specialized equipment.

- Complexity and Cost: Designing and building logarithmic amplifiers with matched transistors can be more complex and costly compared to other amplifier designs. It may involve the use of specialized components and precise manufacturing techniques.

- Component Tolerances: Component tolerances, such as resistor and capacitor tolerances, can introduce variations in the circuit's performance. Choosing high-precision components can help mitigate this limitation but may add to the cost.

In practical practice, designing a logarithmic amplifier with matched transistors is a complex task that comes with various limitations and challenges. We tried to construct the circuit to make it time compensated using simulation software, but designers must carefully consider these limitations and work to mitigate them through precise component selection, calibration, and careful circuit design to achieve the desired level of accuracy and performance for their specific application.